

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 2) - Single Correct

If the point of intersections of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = 4b$, $b > 4$

lie on the curve $y^2 = 3x^2$, then b is equal to:

(1) 12

(2) 5

(3) 6

(4) 10

Q2: 17 March (Shift 2) - Single Correct

Let L be a tangent line to the parabola $y^2 = 4x - 20$ at $(6, 2)$. If L is also a tangent to the ellipse $\frac{x^2}{2} + \frac{y^2}{b} = 1$, then the value of b is equal to :

(1) 11

(2) 14

(3) 16

(4) 20

Answer Key

Q1 (1)

Q2 (2)

Q1 (1)

$$y^2 = 3x^2$$

$$\text{and } x^2 + y^2 = 4b$$

Solve both we get

$$\text{so } x^2 = b$$

$$\frac{x^2}{16} + \frac{3x^2}{b^2} = 1$$

$$\frac{b}{16} + \frac{3}{b} = 1$$

$$b^2 - 16b + 48 = 0$$

$$(b - 12)(b - 4) = 0$$

$$b = 12, b > 4$$

Q2 (2)

Tangent to parabola $2y = 2(x + 6) - 20$

$$\Rightarrow y = x - 4$$

Condition of tangency for ellipse. $16 = 2(1)^2 + b$

$$\Rightarrow b = 14$$